# Affine maps carry Gaussian means and covariances

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$$A \in \mathbb{R}^{m \times d}, \quad b \in \mathbb{R}^m \quad \text{(deterministic)}$$

$$E[x] = \mu, \quad \mathrm{Cov}[x] = \Sigma$$

$$E\left[\sum_{i=1}^{d} x_i^2\right] < \infty$$

## Moment proof

$$\mu_y = E[y] = AE[x] + b = A\mu + b$$

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- Translation shifts the Gaussian cloud's mean without changing its covariance.
- $A$ linearly maps its center and spread; affine closure keeps the transformed cloud Gaussian.